RESEARCH LETTER

# Does the choice of channel attention matter for few-label hyperspectral classification? A controlled comparison under random, standard and patch-disjoint splits

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**ABSTRACT**
Channel-attention blocks are routinely added to lightweight convolutional networks for hyperspectral image classification, and their gains are usually reported on random pixel splits, where overlapping patches leak spatial context into the test set. We ask whether the choice of block still matters under stricter evaluation. Seven modulation blocks (SE, ECA, CBAM, GCT, SRM and two variants of a dispersion-aware block we introduce, spectral-neighbourhood self-modulation, SNSM) are compared on one backbone with one protocol, no test-based model selection, fifteen paired seeds and Holm-corrected Wilcoxon tests. With random 5% splits, blocks differ only on Indian Pines, where SNSM raises average accuracy (AA) by 1.8–3.8 points over SE, ECA, CBAM, GCT and the plain CNN and ties with SRM, the other dispersion-based gate, through classes with one or two training pixels; overall accuracy differs by under 0.2 points. On the standard Indian Pines split all blocks lie within 0.35 points. On patch-disjoint splits of Pavia University and Kennedy Space Center (KSC), accuracy falls by 32–34 points and most block differences are not significant. We also show that the distributed KSC file stores small negative values as unsigned integers; reading it as signed raises seven convolutional variants from 77–92% to over 99% AA.

**KEYWORDS**
Hyperspectral image classification; channel attention; spatial leakage; evaluation protocol; few labels

## 1. Introduction

Hyperspectral images record hundreds of contiguous bands. This makes pixel-level land-cover classification possible, but when labelled samples are few it exposes classifiers to the Hughes phenomenon (Hughes 1968; Bioucas-Dias et al. 2013). Convolutional neural networks (CNNs) that operate on spatial–spectral patches are the usual lightweight choice (Li et al. 2019; Audebert, Le Saux, and Lefevre 2019; Makantasis et al. 2015; Chen et al. 2016; Zhong et al. 2018; Roy et al. 2020), and transformers add global context at a higher cost (Hong et al. 2022; Sun et al. 2022). Channel attention is a cheap way to make such networks input-dependent: squeeze-and-excitation (SE) predicts one scale per channel from pooled means through fully connected (FC) layers (Hu, Shen, and Sun 2018); ECA replaces the FC layers by a one-dimensional

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convolution (Wang et al. 2020); CBAM adds a spatial gate (Woo et al. 2018); GCT normalises channel statistics and gates them with $1 + \tanh(\cdot)$ (Yang et al. 2020); and SRM pools channel means and standard deviations (Lee, Kim, and Nam 2019). Hyperspectral networks have adopted these blocks widely (Mei et al. 2019; Haut et al. 2019; Li et al. 2020; Zhu et al. 2021; Roy et al. 2021).

Most reported gains come from random pixel splits. With patch-based models, training and test patches then overlap, so test accuracy partly measures memorised spatial context (Liang et al. 2017; Nalepa, Myller, and Kawulok 2019). A recent survey documents this leakage (Feng et al. 2023), disjoint sampling changes the accuracy of deep models (Ahmad et al. 2024), and spatially separated splits lower the overall accuracy of CNNs and transformers on Pavia University by about 40 points (Alkhatib 2026). These studies compare architectures. Whether the *choice of channel-attention block* still matters once the protocol is tightened has, to our knowledge, not been tested with paired statistics.

Our contributions are:

- a controlled comparison of seven modulation blocks on one backbone, with identical inputs, schedule and splits, no test-based model selection, fifteen paired seeds and Holm-corrected Wilcoxon tests, under three protocols: random 5% splits, the standard fixed Indian Pines split, and patch-disjoint splits in which no test patch overlaps a training patch (Sections 2–3);
- the finding that block choice matters only in one regime, random splits of Indian Pines whose smallest classes have one or two training pixels, and that the advantage of dispersion-based gates there disappears under the other two protocols (Section 4);
- SNSM, a dispersion-aware block that matches the best baseline, with an ablation showing that the channel variance carries its gain;
- a correction of the Kennedy Space Center (KSC) data encoding, which earlier work treated as sensor saturation (Kakde et al. 2018), with its effect on CNN accuracy.

## 2. Blocks compared

All blocks are inserted in the same backbone. A $3 \times 3$ convolution maps the $B$ input features to $C = 64$ channels (batch normalisation (Ioffe and Szegedy 2015), ReLU). Two residual blocks (He et al. 2016) follow, each computing

$$\mathbf{h} = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{3\times3}(\mathbf{x}))), \qquad \mathbf{y} = \mathrm{ReLU}\big(\mathrm{BN}(\mathrm{Conv}_{3\times3}(\mathcal{M}(\mathbf{h}))) + \mathbf{x}\big), \quad (1)$$

where $\mathcal{M}$ is the block under study ($\mathcal{M}$ = identity for the plain CNN). Global average pooling, dropout (0.4) (Srivastava et al. 2014) and a linear layer produce the class scores. SE, ECA (kernel 3, the adaptive value for 64 channels) and CBAM (reduction 16, $7 \times 7$ spatial gate) output a sigmoid scale per channel; GCT uses its $\ell_2$ embedding and an identity start; SRM gates each channel from its mean and standard deviation through a channel-wise FC layer and batch normalisation.

**SNSM.** For $\mathbf{h} \in \mathbb{R}^{C\times H\times W}$, SNSM computes for every channel $c$ the spatial mean $\mu_c$, the spatial variance $v_c$ and the cross-channel gradient $g_c = (\mu_{c+1} - \mu_{c-1})/2$. Each descriptor vector is standardised across channels, the three are stacked into $\mathbf{S} \in \mathbb{R}^{3\times C}$, and two one-dimensional convolutions along the channel axis (8 hidden channels, kernel

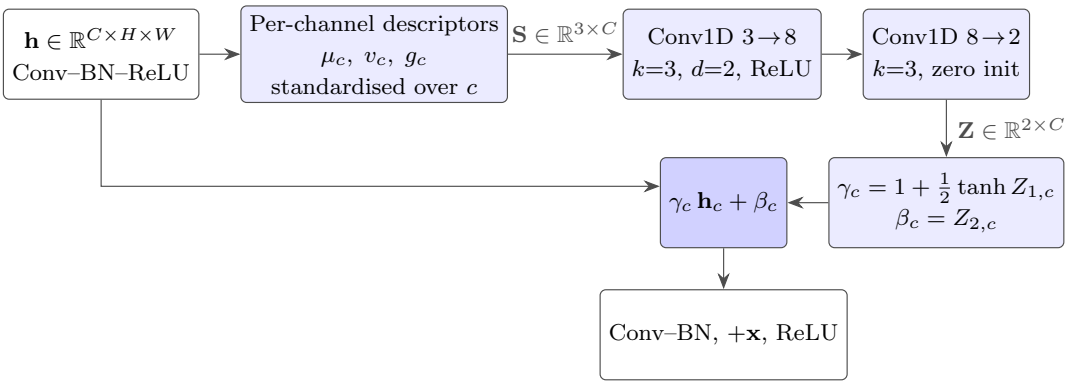



**Figure 1.** SNSM inside a residual block. Per-channel descriptors of the feature map $\mathbf{h}$ are filtered along the channel axis by two one-dimensional convolutions, whose output sets a bounded scale $\gamma_c$ and a shift $\beta_c$ for each channel. The second convolution starts at zero, so the block is initially the identity.

3, dilation 2 in the first, replicate padding) give $\mathbf{Z} \in \mathbb{R}^{2\times C}$. The modulation is

$$\gamma_c = 1 + \tfrac{1}{2}\tanh(Z_{1,c}), \qquad \beta_c = Z_{2,c}, \qquad \mathcal{M}(\mathbf{h})_c = \gamma_c\,\mathbf{h}_c + \beta_c. \tag{2}$$

The second convolution starts at zero, so the block is initially the identity; it adds 130 parameters per block (Figure 1). SNSM ($\gamma$) drops the shift. Like SRM it uses channel dispersion; unlike SRM it filters the descriptors jointly across neighbouring channels and can shift as well as scale. Because learned channels have no wavelength order, "neighbourhood" refers to adjacent feature channels.

## 3. Data and protocols

**Data.** We use Indian Pines (IP), Pavia University (PU), KSC and Botswana (BW) (Baumgardner, Biehl, and Landgrebe 2015; Gamba 2004; Grupo de Inteligencia Computacional 2020) (Table 1). Each band is smoothed with a $5\times5$ Wiener filter, principal component analysis (PCA) (Jolliffe and Cadima 2016), fitted without labels on all pixels, keeps 30 components, and the db4 wavelet approximation of each component (Mallat 1989) is resized and appended, giving $B = 60$ features. Each labelled pixel is represented by its $15 \times 15 \times B$ patch.

**KSC encoding.** The commonly distributed KSC file is stored as unsigned 16-bit integers, but it contains small negative values ($-27$ to $-1$) that are stored in two's complement and are therefore read as 65,509–65,535; about 5% of its pixels contain at least one such value. Kakde et al. (2018) noticed these values, described them as saturation and set them to zero, but widely used toolboxes still load the file unchanged and scale it by its global minimum and maximum (Audebert, Le Saux, and Lefevre 2019), which compresses the valid range of 0–1,244 into less than 2% of the scale. We read KSC as signed integers. Without this correction, the wrapped values dominate the PCA and the seven CNN variants reach only 77–92% AA on KSC with random 5% splits, instead of 99.2–99.5%.

**Protocols.** *Random*: for each class, 5% of the labelled pixels (at least one) are drawn for training and the rest are used for testing, giving 505/9,744 (IP), 2,135/40,641 (PU), 256/4,955 (KSC) and 156/3,092 (BW) pixels; three IP classes then have one or two training pixels. *Standard*: the fixed IP training and test sets distributed with SpectralFormer (Hong et al. 2022) (695/9,671 pixels; 15–50 training pixels per class).

**Table 1.** Benchmark scenes and the random 5% splits (seed 42; split sizes are the same for every seed).

| Scene | Sensor | Size (pixels) | Bands | GSD (m) | Classes | Labelled | Train/test (random) |
|---|---|---|---|---|---|---|---|
| Indian Pines (IP) | AVIRIS | $145 \times 145$ | 200 | 20 | 16 | 10,249 | 505/9,744 |
| Pavia University (PU) | ROSIS | $610 \times 340$ | 103 | 1.3 | 9 | 42,776 | 2,135/40,641 |
| Kennedy Space Center (KSC) | AVIRIS | $512 \times 614$ | 176 | 18 | 13 | 5,211 | 256/4,955 |
| Botswana (BW) | Hyperion | $1476 \times 256$ | 145 | 30 | 14 | 3,248 | 156/3,092 |




**Figure 2.** Patch-disjoint split of Pavia University (seed 42). Training pixels form one compact cluster per class; test pixels whose $15 \times 15$ patch would overlap a training patch are excluded.

Its training pixels come from separate regions, but with $15 \times 15$ patches every test patch still overlaps a training patch. The corresponding Pavia University and Houston sets were no longer available from the authors' link. *Patch-disjoint* (PU, KSC): for each class, the 5% of its pixels nearest a random centre form a compact training cluster, and every test pixel within 14 pixels of any training pixel is discarded, so no test patch overlaps a training patch (Figure 2); 33,847–36,453 (PU) and 3,077–3,513 (KSC) test pixels remain, covering all 9 PU classes and at least 11 of the 13 KSC classes. The same rule leaves no test pixels for five IP classes, so we use the standard split for IP.

**Training and statistics.** Every model sees the same features and splits for a given seed. Training uses cross-entropy, AdamW (Loshchilov and Hutter 2019) (learning rate $10^{-3}$, weight decay $10^{-4}$), step decay (factor 0.5 every 40 epochs), batch size 64 and exactly 100 epochs, without augmentation or early stopping; the final model is evaluated once, so no validation or test data are used for selection. We use 15 seeds (42–56) for all eight backbone variants on IP (random and standard) and on the patch-disjoint splits, and five seeds elsewhere. We compare SNSM ($\gamma$+$\beta$) with each other block by paired same-seed differences and two-sided Wilcoxon signed-rank tests, Holm-corrected over the seven comparisons for each scene and protocol. Runs used PyTorch (Paszke et al. 2019) on NVIDIA Tesla T4 GPUs, except ten KSC patch-disjoint seeds,

**Table 2.** Random 5% splits: OA and AA (%, mean ± std). The eight backbone variants use 15 seeds on IP and five elsewhere; FC ($\gamma$+$\beta$), HybridSN (Roy et al. 2020) and SSFTT (Sun et al. 2022) use five. Best mean per column in bold.

| Method | IP | | KSC | | PU | | BW | |
|---|---|---|---|---|---|---|---|---|
| | OA | AA | OA | AA | OA | AA | OA | AA |
| Plain CNN | 96.28±0.79 | 89.49±2.44 | 99.65±0.34 | 99.51±0.36 | **99.59±0.07** | 99.01±0.16 | 99.09±0.55 | 98.87±0.90 |
| SE ($\gamma$) | 96.34±0.62 | 90.04±2.18 | 99.61±0.28 | 99.43±0.38 | 99.53±0.19 | 98.80±0.56 | 98.95±0.82 | 98.76±1.07 |
| ECA ($\gamma$) | 96.28±0.68 | 90.87±2.53 | 99.61±0.47 | 99.40±0.66 | 99.58±0.10 | 99.00±0.20 | **99.13±0.76** | **98.90±1.08** |
| CBAM | 96.19±0.60 | 91.58±1.49 | 99.54±0.56 | 99.23±0.83 | 99.58±0.06 | **99.11±0.18** | 98.99±0.87 | 98.78±1.20 |
| GCT | 96.28±0.80 | 90.07±2.34 | **99.69±0.26** | **99.55±0.28** | 99.58±0.08 | 98.99±0.18 | 99.11±0.56 | 98.88±0.91 |
| SRM | 96.20±0.71 | 93.04±2.14 | 99.69±0.36 | 99.51±0.49 | 99.55±0.07 | 98.92±0.17 | 98.87±0.63 | 98.67±0.97 |
| SNSM ($\gamma$) | 96.17±0.73 | 92.89±2.00 | 99.69±0.26 | 99.50±0.39 | 99.49±0.05 | 98.91±0.16 | 99.10±0.62 | 98.89±0.96 |
| SNSM ($\gamma$+$\beta$) | 96.27±0.67 | **93.33±1.68** | 99.63±0.55 | 99.45±0.73 | 99.47±0.05 | 98.93±0.16 | 99.04±0.69 | 98.85±1.03 |
| FC ($\gamma$+$\beta$) | 96.24±0.52 | 92.51±1.60 | 99.63±0.40 | 99.48±0.47 | 99.51±0.06 | 98.88±0.14 | 98.80±0.70 | 98.63±0.98 |
| HybridSN | 88.30±4.91 | 73.64±8.28 | 95.00±3.02 | 91.20±5.32 | 97.84±1.52 | 95.46±4.12 | 77.16±37.92 | 75.56±38.31 |
| SSFTT | **96.71±0.79** | 92.35±1.81 | 99.38±0.28 | 99.15±0.32 | – | – | – | – |

**Table 3.** AA under the random protocol (R) and OA and AA under the held-out protocol (H): the standard split for IP and the patch-disjoint split for PU and KSC (%, mean ± std; 15 seeds except $AA_R$ on PU and KSC, five seeds). The last row is the range of the eight block means.

| Block | IP | | | PU | | | KSC | | |
|---|---|---|---|---|---|---|---|---|---|
| | $AA_R$ | $OA_H$ | $AA_H$ | $AA_R$ | $OA_H$ | $AA_H$ | $AA_R$ | $OA_H$ | $AA_H$ |
| Plain CNN | 89.49±2.44 | 95.15±0.29 | 97.24±0.23 | 99.01±0.16 | 55.31±9.82 | 67.03±5.03 | 99.51±0.36 | 75.13±4.97 | 63.55±6.41 |
| SE ($\gamma$) | 90.04±2.18 | 94.88±0.46 | 97.25±0.26 | 98.80±0.56 | 55.91±9.73 | **67.81±4.39** | 99.43±0.38 | 76.79±4.44 | **67.20±6.66** |
| ECA ($\gamma$) | 90.87±2.53 | 94.92±0.33 | 97.15±0.22 | 99.00±0.20 | 56.30±9.17 | 67.73±4.43 | 99.40±0.66 | 76.40±4.81 | 65.15±5.86 |
| CBAM | 91.58±1.49 | 95.08±0.67 | 97.45±0.29 | **99.11±0.18** | 56.11±10.31 | 66.31±4.80 | 99.23±0.83 | 76.68±4.74 | 66.10±7.39 |
| GCT | 90.07±2.34 | 95.08±0.26 | 97.23±0.22 | 98.99±0.18 | 54.59±9.68 | 66.42±4.57 | **99.55±0.28** | 75.20±5.16 | 63.49±6.80 |
| SRM | 93.04±2.14 | 95.20±0.54 | **97.50±0.22** | 98.92±0.17 | 56.02±9.99 | 66.00±4.73 | 99.51±0.49 | 74.36±4.22 | 62.75±6.38 |
| SNSM ($\gamma$) | 92.89±2.00 | 95.07±0.52 | 97.37±0.22 | 98.91±0.16 | 55.41±9.44 | 67.31±4.57 | 99.50±0.39 | 76.86±4.19 | 65.60±7.15 |
| SNSM ($\gamma$+$\beta$) | **93.33±1.68** | **95.23±0.42** | 97.48±0.26 | 98.93±0.16 | **56.80±11.83** | 67.34±4.95 | 99.45±0.73 | **77.11±3.54** | 67.01±6.00 |
| Spread of block means | 3.84 | 0.34 | 0.35 | 0.31 | 2.21 | 1.80 | 0.32 | 2.75 | 4.44 |

which ran on an RTX 3050 laptop GPU.

## 4. Results

**Random splits.** On PU, KSC and BW the eight backbone variants lie within 0.32 AA points of each other (Table 2). Only IP separates them: SNSM ($\gamma$+$\beta$) has the highest AA, and it is significantly better than the plain CNN, SE, ECA, CBAM and GCT (by 1.75–3.84 points; Table 4, first column), but not than SRM or SNSM ($\gamma$). The overall accuracy of all eight variants differs by only 0.17 points. The gain therefore comes from a few classes: Oats and Grass-pasture-mowed have one training pixel and 19 and 27 test pixels, so a change of a few test pixels moves their per-class accuracy by tens of points. The two blocks that see channel dispersion, SRM and SNSM, are the two best and are statistically tied. The externally designed HybridSN and SSFTT, trained under our protocol, are less accurate than the backbone variants.

**Standard and patch-disjoint splits.** With the standard IP split, every class has at least 15 training pixels and all eight blocks reach 97.15–97.50% AA (Table 3); the only significant difference left, against ECA, is 0.33 points. The 3.84-point spread of the random protocol thus does not survive a split that gives the small classes enough training pixels. On the patch-disjoint splits accuracy collapses: AA falls by 32.0 points on PU and 34.3 on KSC, averaged over blocks, and OA falls to 55–57% and 74–77%. Block means now spread by 1.80 (PU) and 4.44 (KSC) points, but the seed-to-seed standard deviation is 4.4–7.4 points. On PU no difference is significant; on KSC, SNSM ($\gamma$+$\beta$) is significantly better than the plain CNN, GCT and SRM, but not than SE, ECA, CBAM or SNSM ($\gamma$), and SE has the highest mean AA (Figure 3). The ranking of blocks also changes between protocols: SRM, tied best on random IP, is the lowest on patch-disjoint KSC.

**What drives the gain on random IP.** Removing descriptors from SNSM on

**Table 4.** Paired AA differences SNSM ($\gamma$+$\beta$) − block (points; seeds won in parentheses). *: Holm-corrected two-sided Wilcoxon $p < 0.05$.

| SNSM ($\gamma$+$\beta$) − | IP, random | IP, standard | PU, disjoint | KSC, disjoint |
|---|---|---|---|---|
| Plain CNN | +3.84* (14/15) | +0.24 (11/15) | +0.31 (8/15) | +3.46* (11/15) |
| SE ($\gamma$) | +3.29* (15/15) | +0.23 (11/15) | -0.47 (7/15) | -0.19 (8/15) |
| ECA ($\gamma$) | +2.45* (11/15) | +0.33* (13/15) | -0.39 (5/15) | +1.86 (11/15) |
| CBAM | +1.75* (13/15) | +0.04 (7/15) | +1.03 (10/15) | +0.91 (7/15) |
| GCT | +3.26* (14/15) | +0.25 (11/15) | +0.92 (9/15) | +3.51* (12/15) |
| SRM | +0.29 (8/15) | -0.02 (8/15) | +1.33 (11/15) | +4.25* (12/15) |
| SNSM ($\gamma$) | +0.44 (9/15) | +0.12 (10/15) | +0.03 (8/15) | +1.40 (9/15) |

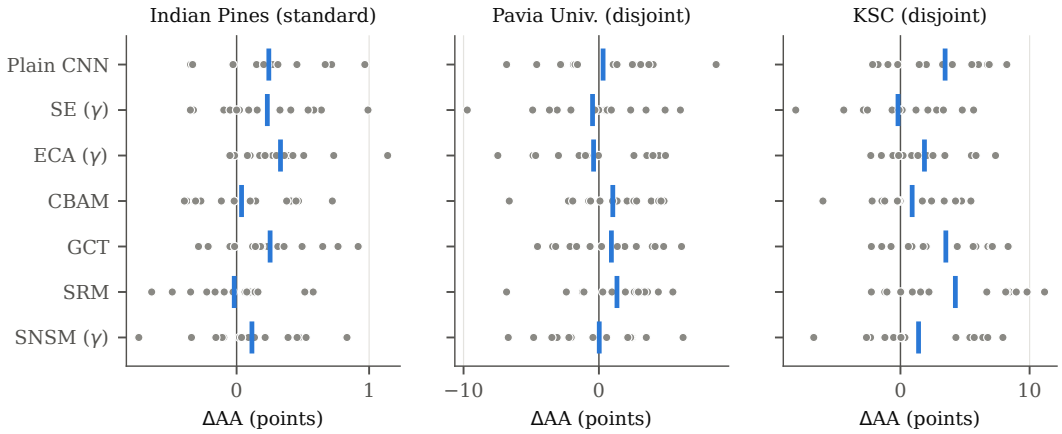



**Figure 3.** Paired same-seed AA differences SNSM ($\gamma$+$\beta$) − block under the held-out protocols. Grey dots are the 15 seeds; the blue bar is their mean.

random IP (five seeds; Table 5) shows that the variance carries the gain: with the mean alone AA falls by 1.16 points, and adding the gradient to the mean does not recover it; without the zero initialisation AA is lower on all five seeds. This matches the tie between SNSM and SRM, which also gates on dispersion. When SNSM is applied to the raw bands in wavelength order instead of to feature channels, it also raises IP AA by 3.9–4.8 points over SE, ECA and GCT applied the same way (five seeds, five wins each), but this raw-band pipeline is weaker overall and ties on PU and KSC (results in the repository).

**Cost.** The channel blocks add between 6 (ECA) and 4,256 (SE) parameters to the plain CNN, and SNSM adds 260 (Table 6). SNSM's descriptor computation consists of many small operations, so its batch-1 latency is about 3.7 times and its training time about 1.7 times that of the plain CNN.

## 5. Discussion and limitations

The choice of channel-attention block changed AA by more than half a point only when the smallest classes had one or two training pixels and test patches overlapped training patches. In that regime, the two gates that see channel dispersion (SRM, SNSM) were clearly better than SE, ECA, CBAM and GCT. Once the small classes had enough training pixels, or once no test patch overlapped a training patch, the blocks were practically or statistically indistinguishable, while the protocol itself moved AA by more than 30 points. Gains of a few AA points reported for a new block on a random

**Table 5.** SNSM ($\gamma$+$\beta$) ablation on IP, random 5% splits (AA, %; five seeds). $\Delta$AA is the paired difference from the full block; wins count seeds on which the variant is better.

| Descriptors | AA | $\Delta$AA (wins/5) |
|---|---|---|
| mean, var, grad (SNSM) | 94.16±1.07 | – |
| mean | 93.01±0.72 | -1.16 (1/5) |
| mean, var | 94.55±1.67 | +0.39 (4/5) |
| mean, grad | 93.37±1.09 | -0.80 (1/5) |
| mean, var, grad; default init. | 92.67±1.15 | -1.49 (0/5) |

**Table 6.** Model size and cost for a $15 \times 15 \times 60$ IP patch. Training time is the mean per run (100 epochs, random splits).

| Method | Params (M) | MMACs[a] | Latency (ms)[b] | Train IP (s) | Train PU (s) |
|---|---|---|---|---|---|
| Plain CNN | 0.184 | 41.2 | 0.69 | 4.2 | 17.6 |
| SE ($\gamma$) | 0.188 | 41.2 | 1.17 | 4.5 | 19.3 |
| ECA ($\gamma$) | 0.184 | 41.2 | 0.98 | 4.8 | 19.8 |
| CBAM | 0.185 | 41.3 | 1.55 | 5.9 | 24.9 |
| GCT | 0.184 | 41.2 | 1.11 | 4.7 | 20.0 |
| SRM | 0.184 | 41.2 | 1.22 | 4.8 | 20.2 |
| SNSM ($\gamma$) | 0.184 | 41.3 | 2.54 | 7.1 | 29.5 |
| SNSM ($\gamma$+$\beta$) | 0.184 | 41.3 | 2.54 | 7.2 | 30.2 |
| FC ($\gamma$+$\beta$) | 0.190 | 41.2 | 1.31 | 4.7 | 19.8 |
| HybridSN | 1.743 | 137.4 | 0.71 | 26.2 | 115.3 |
| SSFTT | 0.291 | 34.9 | 1.25 | 7.6 | – |

[a] Multiply–accumulates of convolution and linear layers counted by `thop`; tensor operations such as attention products and descriptor statistics are not included. [b] Batch size 1 on a Tesla T4.

split should therefore not be read as evidence of better generalisation.

Several limitations apply. The standard IP split still lets test patches overlap training patches, and we could not obtain the standard PU and Houston splits; our patch-disjoint construction is one of several possible and leaves few test pixels for some KSC classes. PCA is fitted on all pixels of each scene without labels. All blocks share one backbone, input pipeline and schedule, none was tuned to it, and HybridSN and SSFTT are less accurate here than in their original papers. Wilcoxon tests with 15 seeds detect consistent shifts but not their practical relevance, so we report effect sizes alongside them.

## 6. Conclusion

Under one strict protocol, the choice among seven lightweight channel-attention blocks matters little for few-label hyperspectral classification. Dispersion-based gates, including the proposed SNSM, add 2–4 AA points on random Indian Pines splits through classes with one or two training pixels, but this advantage vanishes on the standard split and is mostly not significant on patch-disjoint splits, where accuracy drops by 32–34 points for every block. Evaluation protocol and data handling, such as the KSC encoding corrected here, affect reported accuracy far more than the attention block.

## Disclosure statement

The authors report there are no competing interests to declare.

## Funding

No external funding was received for this research.

## Data and code availability

The datasets are publicly available (Grupo de Inteligencia Computacional 2020); the standard IP split is distributed with SpectralFormer (Hong et al. 2022). Code, per-run results and the scripts that generate every table and figure are available at `https://github.com/Devguru-codes/research_paper_code`.

## Use of generative AI

Generative AI (Claude, Anthropic) was used to assist with writing experiment code, checking references and editing the manuscript text. The authors reviewed and verified all code, results and text, and take full responsibility for the content.

## References

Ahmad, Muhammad, Manuel Mazzara, Salvatore Distefano, Adil Mehmood Khan, and Hamad Ahmed Altuwaijri. 2024. "Improving Generalization for Hyperspectral Image Classification: The Impact of Disjoint Sampling on Deep Models." *Computers, Materials & Continua* 81 (1): 503–532. https://doi.org/10.32604/cmc.2024.056318.

Alkhatib, Mohammed Q. 2026. "Data Leakage in Patch-Based Hyperspectral Image Classification: Quantifying the Impact of Spatial Overlap." arXiv preprint arXiv:2610.08770. https://doi.org/10.48550/arXiv.2610.08770.

Audebert, Nicolas, Bertrand Le Saux, and Sebastien Lefevre. 2019. "Deep Learning for Classification of Hyperspectral Data: A Comparative Review." *IEEE Geoscience and Remote Sensing Magazine* 7 (2): 159–173. https://doi.org/10.1109/MGRS.2019.2912563.

Baumgardner, Marion F., Larry L. Biehl, and David A. Landgrebe. 2015. "220 Band AVIRIS Hyperspectral Image Data Set: June 12, 1992 Indian Pine Test Site 3." Purdue University Research Repository. https://doi.org/10.4231/R7RX991C.

Bioucas-Dias, Jose M., Antonio Plaza, Gustavo Camps-Valls, Paul Scheunders, Nasser Nasrabadi, and Jocelyn Chanussot. 2013. "Hyperspectral Remote Sensing Data Analysis and Future Challenges." *IEEE Geoscience and Remote Sensing Magazine* 1 (2): 6–36. https://doi.org/10.1109/MGRS.2013.2244672.

Chen, Yushi, Hanlu Jiang, Chunyang Li, Xiuping Jia, and Pedram Ghamisi. 2016. "Deep Feature Extraction and Classification of Hyperspectral Images Based on Convolutional Neural Networks." *IEEE Transactions on Geoscience and Remote Sensing* 54 (10): 6232–6251. https://doi.org/10.1109/TGRS.2016.2584107.

Feng, Hao, Yongcheng Wang, Zheng Li, Ning Zhang, Yuxi Zhang, and Yunxiao Gao. 2023. "Information Leakage in Deep Learning-Based Hyperspectral Image Classification: A Survey." *Remote Sensing* 15 (15): 3793. https://doi.org/10.3390/rs15153793.

Gamba, Paolo. 2004. "ROSIS Pavia University Hyperspectral Data Set." Telecommunications and Remote Sensing Laboratory, University of Pavia. `https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes`.

Grupo de Inteligencia Computacional. 2020. "Hyperspectral Remote Sensing Scenes." University of the Basque Country (UPV/EHU). `https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes`.

Haut, Juan Mario, Mercedes E. Paoletti, Javier Plaza, Antonio Plaza, and Jun Li. 2019. "Visual Attention-Driven Hyperspectral Image Classification." *IEEE Transactions on Geoscience and Remote Sensing* 57 (10): 8065–8080. https://doi.org/10.1109/TGRS.2019.2918080.

He, Kaiming, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. "Deep Residual Learning for Image Recognition." In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778. https://doi.org/10.1109/CVPR.2016.90.

Hong, Danfeng, Zhu Han, Jing Yao, Lianru Gao, Bing Zhang, Antonio Plaza, and Jocelyn Chanussot. 2022. "SpectralFormer: Rethinking Hyperspectral Image Classification With Transformers." *IEEE Transactions on Geoscience and Remote Sensing* 60: 1–15. https://doi.org/10.1109/TGRS.2021.3130716.

Hu, Jie, Li Shen, and Gang Sun. 2018. "Squeeze-and-Excitation Networks." In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 7132–7141. https://doi.org/10.1109/CVPR.2018.00745.

Hughes, G. 1968. "On the mean accuracy of statistical pattern recognizers." *IEEE Transactions on Information Theory* 14 (1): 55–63. https://doi.org/10.1109/TIT.1968.1054102.

Ioffe, Sergey, and Christian Szegedy. 2015. "Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift." In *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, 448–456. https://doi.org/10.48550/arXiv.1502.03167.

Jolliffe, Ian T., and Jorge Cadima. 2016. "Principal component analysis: a review and recent developments." *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 374 (2065): 20150202. https://doi.org/10.1098/rsta.2015.0202.

Kakde, Deovrat, Arin Chaudhuri, Hansi Jiang, Carol Sadek, Seunghyun Kong, and Yuwei Liao. 2018. "A New Bandwidth Selection Criterion for Using SVDD to Analyze Hyperspectral Data." In *Algorithms and Technologies for Multispectral, Hyperspectral, and Ultraspectral Imagery XXIV*, Proc. SPIE. https://doi.org/10.1117/12.2314964.

Lee, Hyunjae, Hyo-Eun Kim, and Hyeonseob Nam. 2019. "SRM: A Style-Based Recalibration Module for Convolutional Neural Networks." In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, 1854–1862. https://doi.org/10.1109/ICCV.2019.00194.

Li, Rui, Shunyi Zheng, Chenxi Duan, Yang Yang, and Xiqi Wang. 2020. "Classification of Hyperspectral Image Based on Double-Branch Dual-Attention Mechanism Network." *Remote Sensing* 12 (3): 582. https://doi.org/10.3390/rs12030582.

Li, Shutao, Weiwei Song, Leyuan Fang, Yushi Chen, Pedram Ghamisi, and Jon Atli Benediktsson. 2019. "Deep Learning for Hyperspectral Image Classification: An Overview." *IEEE Transactions on Geoscience and Remote Sensing* 57 (9): 6690–6709. https://doi.org/10.1109/TGRS.2019.2907932.

Liang, Jie, Jun Zhou, Yuntao Qian, Lian Wen, Xiao Bai, and Yongsheng Gao. 2017. "On the Sampling Strategy for Evaluation of Spectral-Spatial Methods in Hyperspectral Image Classification." *IEEE Transactions on Geoscience and Remote Sensing* 55 (2): 862–880. https://doi.org/10.1109/TGRS.2016.2616489.

Loshchilov, Ilya, and Frank Hutter. 2019. "Decoupled Weight Decay Regularization." In *International Conference on Learning Representations (ICLR)*, https://doi.org/10.48550/arXiv.1711.05101.

Makantasis, Konstantinos, Konstantinos Karantzalos, Anastasios Doulamis, and Nikolaos Doulamis. 2015. "Deep supervised learning for hyperspectral data classification through convolutional neural networks." In *2015 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 4959–4962. https://doi.org/10.1109/IGARSS.2015.7326945.

Mallat, S.G. 1989. "A theory for multiresolution signal decomposition: the wavelet representation." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 11 (7): 674–693. https://doi.org/10.1109/34.192463.

Mei, Xiaoguang, Erting Pan, Yong Ma, Xiaobing Dai, Jun Huang, Fan Fan, Qinglei Du, Hong Zheng, and Jiayi Ma. 2019. "Spectral-Spatial Attention Networks for Hyperspectral Image Classification." *Remote Sensing* 11 (8): 963. https://doi.org/10.3390/rs11080963.

Nalepa, Jakub, Michal Myller, and Michal Kawulok. 2019. "Validating Hyperspectral Im-

age Segmentation." *IEEE Geoscience and Remote Sensing Letters* 16 (8): 1264–1268. https://doi.org/10.1109/LGRS.2019.2895697.

Paszke, Adam, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, et al. 2019. "PyTorch: An Imperative Style, High-Performance Deep Learning Library." In *Advances in Neural Information Processing Systems 32 (NeurIPS)*, 8024–8035. https://doi.org/10.48550/arXiv.1912.01703.

Roy, Swalpa Kumar, Gopal Krishna, Shiv Ram Dubey, and Bidyut B. Chaudhuri. 2020. "HybridSN: Exploring 3-D–2-D CNN Feature Hierarchy for Hyperspectral Image Classification." *IEEE Geoscience and Remote Sensing Letters* 17 (2): 277–281. https://doi.org/10.1109/LGRS.2019.2918719.

Roy, Swalpa Kumar, Suvojit Manna, Tiecheng Song, and Lorenzo Bruzzone. 2021. "Attention-Based Adaptive Spectral–Spatial Kernel ResNet for Hyperspectral Image Classification." *IEEE Transactions on Geoscience and Remote Sensing* 59 (9): 7831–7843. https://doi.org/10.1109/TGRS.2020.3043267.

Srivastava, Nitish, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. 2014. "Dropout: A Simple Way to Prevent Neural Networks from Overfitting." *Journal of Machine Learning Research* 15 (56): 1929–1958. `https://jmlr.org/papers/v15/srivastava14a.html`.

Sun, Le, Guangrui Zhao, Yuhui Zheng, and Zebin Wu. 2022. "Spectral–Spatial Feature Tokenization Transformer for Hyperspectral Image Classification." *IEEE Transactions on Geoscience and Remote Sensing* 60: 1–14. https://doi.org/10.1109/TGRS.2022.3144158.

Wang, Qilong, Banggu Wu, Pengfei Zhu, Peihua Li, Wangmeng Zuo, and Qinghua Hu. 2020. "ECA-Net: Efficient Channel Attention for Deep Convolutional Neural Networks." In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 11531–11539. https://doi.org/10.1109/CVPR42600.2020.01155.

Woo, Sanghyun, Jongchan Park, Joon-Young Lee, and In So Kweon. 2018. "CBAM: Convolutional Block Attention Module." In *Computer Vision – ECCV 2018, Lecture Notes in Computer Science, vol. 11211*, 3–19. https://doi.org/10.1007/978-3-030-01234-2_1.

Yang, Zongxin, Linchao Zhu, Yu Wu, and Yi Yang. 2020. "Gated Channel Transformation for Visual Recognition." In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 11791–11800. https://doi.org/10.1109/CVPR42600.2020.01181.

Zhong, Zilong, Jonathan Li, Zhiming Luo, and Michael Chapman. 2018. "Spectral–Spatial Residual Network for Hyperspectral Image Classification: A 3-D Deep Learning Framework." *IEEE Transactions on Geoscience and Remote Sensing* 56 (2): 847–858. https://doi.org/10.1109/TGRS.2017.2755542.

Zhu, Minghao, Licheng Jiao, Fang Liu, Shuyuan Yang, and Jianing Wang. 2021. "Residual Spectral–Spatial Attention Network for Hyperspectral Image Classification." *IEEE Transactions on Geoscience and Remote Sensing* 59 (1): 449–462. https://doi.org/10.1109/TGRS.2020.2994057.